S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 3

Aim: To implement **Logistic Regression** for binary or multiclass classification problems and analyze the classification performance.

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USN NO.	**CS23071**
Semester/Year	**06/03**
DATE OF PERFORMANCE	**24/01/2026**
DATE OF SUBMISSION	**31/01/2026**

AIM: To implement Logistic Regression for binary or multiclass classification problems and analyze the classification performance.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand the use of odds, odds ratios and transformations in logistic regression.
- Logistic regression is a statistical analysis method to predict a binary outcome
- To measure the relationship between a categorical dependent variable and one or more independent variables (usually continuous) by plotting the dependent variables' probability scores.
- Able to calculate both simple and multiple regression models. You will learn how to assess the model's "fit", test model assumptions, and transform predictor and response variables to improve outcomes

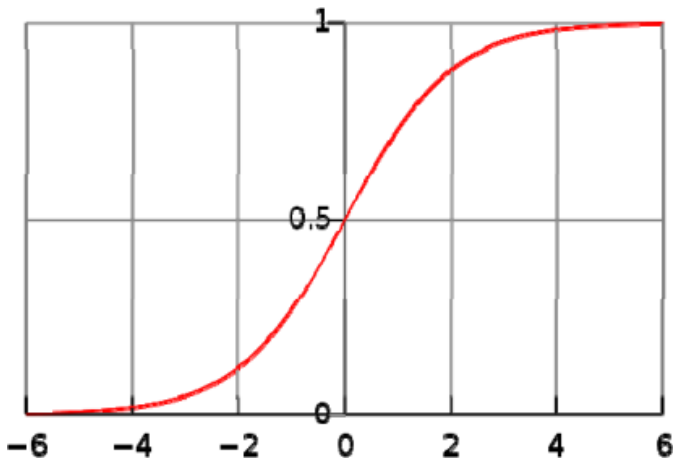
THEORY:

Logistic regression

- Name is somewhat misleading. Really a technique for classification, not regression. technique for classification, not regression.
- "Regression" comes from fact that we fit a linear model to the feature space
- Involves a more probabilistic view of classification

$$p = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}} \qquad \text{logistic function}$$

Standard logistic function



Using a logistic regression model

Can interpret prediction from a logistic regression model as:

- A probability of class membership –
- A class assignment by applying threshold to A class assignment, by applying threshold to probability.
- Threshold represents decision boundary in feature space

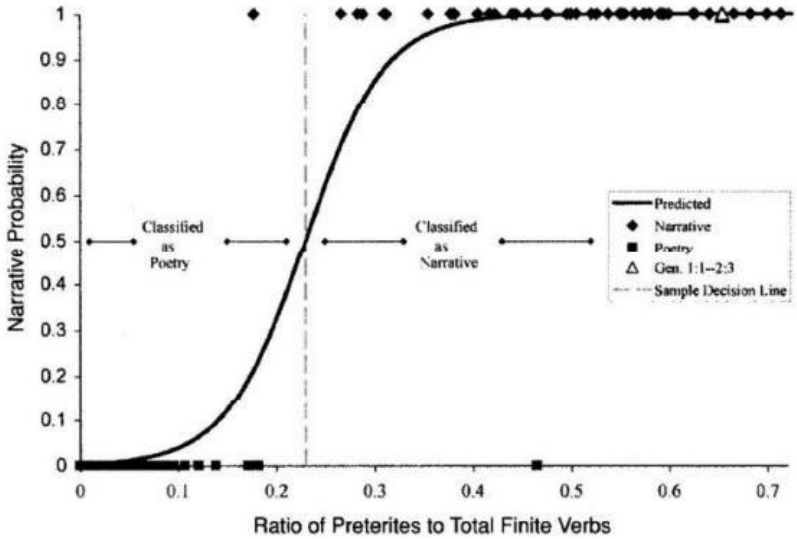
Training a logistic regression model

Need to optimize β so the model gives the best possible reproduction of training set labels possible reproduction of training set labels

– Usually done by numerical approximation of maximum likelihood

– On really large datasets, may use stochastic gradient descent

Logistic regression in one dimension

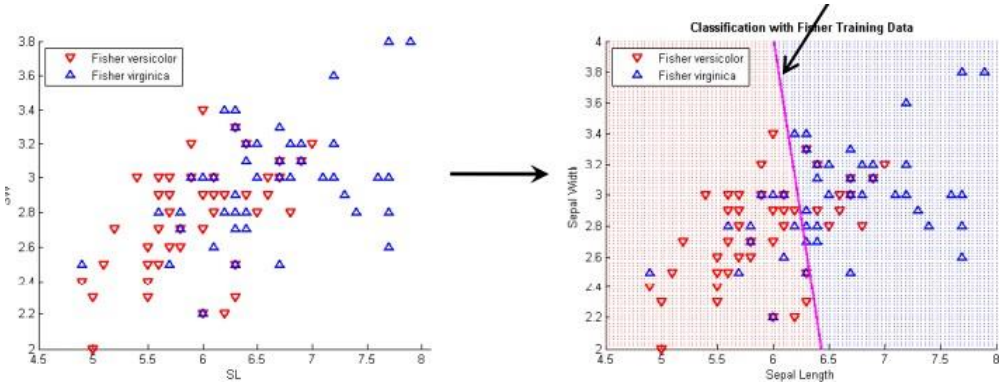


Logistic regression in two dimensions

Subset of Fisher iris dataset

– Two classes –

First two columns (SL, SW)



STEPS:

1) Split the data set into training and test sets.

2) Fit logistic regression model on training set.

3) Prepare Confusion matrix.

4) Compute model accuracy, precision, recall.

PROGRAM CODE:

OUTPUT (SCREENSHOT):

Abhishek.ipynb

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Name:Abhishek Yadav USN NO:CS23071 Aim:To implement Logistic Regression for binary classification problems and analyze the classification performance.

[2]
```
import seaborn as sns
import pandas as pd
import numpy as np
```

[3]
```
df= sns.load_dataset('iris')
df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

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Abhishek.ipynb

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[4]
```
df.describe()
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

[9]
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
```

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```
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

[11]

```
df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with df New interactive sheet

[10]

```
df
```

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[10]

```
df
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

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[17]
```
df=df[df['species']!='setosa']
```

[18]
```
df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
50	7.0	3.2	4.7	1.4	versicolor
51	6.4	3.2	4.5	1.5	versicolor
52	6.9	3.1	4.9	1.5	versicolor
53	5.5	2.3	4.0	1.3	versicolor
54	6.5	2.8	4.6	1.5	versicolor

Next steps: Generate code with df New interactive sheet

[20]
```
df['species']=df['species'].map({'versicolor':0,'virginica':1})
```

[21]
```
df.head()
```

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51	6.4	3.2	4.5	1.5	NaN
52	6.9	3.1	4.9	1.5	NaN
53	5.5	2.3	4.0	1.3	NaN
54	6.5	2.8	4.6	1.5	NaN

Next steps: Generate code with df New interactive sheet

[22]
```
X=df.iloc[:,:-1]
print(X)
```

```
     sepal_length  sepal_width  petal_length  petal_width
50           7.0          3.2           4.7          1.4
51           6.4          3.2           4.5          1.5
52           6.9          3.1           4.9          1.5
53           5.5          2.3           4.0          1.3
54           6.5          2.8           4.6          1.5
..           ...          ...           ...          ...
145          6.7          3.0           5.2          2.3
146          6.3          2.5           5.0          1.9
147          6.5          3.0           5.2          2.0
148          6.2          3.4           5.4          2.3
```

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[23]
```python
y=df.iloc[:,:-1]
print(y)
```

```
     sepal_length  sepal_width  petal_length  petal_width
50            7.0          3.2           4.7          1.4
51            6.4          3.2           4.5          1.5
52            6.9          3.1           4.9          1.5
53            5.5          2.3           4.0          1.3
54            6.5          2.8           4.6          1.5
..            ...          ...           ...          ...
145           6.7          3.0           5.2          2.3
146           6.3          2.5           5.0          1.9
147           6.5          3.0           5.2          2.0
148           6.2          3.4           5.4          2.3
149           5.9          3.0           5.1          1.8

[100 rows x 4 columns]
```

[25]
```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=0.7,test_size = 0.3,random_state = 42)
```

[27]
```python
from sklearn.linear_model import LogisticRegression
classifier= LogisticRegression()
```

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Abhishek.ipynb

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```
51            6.4          3.2           4.5          1.5
52            6.9          3.1           4.9          1.5
53            5.5          2.3           4.0          1.3
54            6.5          2.8           4.6          1.5
..            ...          ...           ...          ...
145           6.7          3.0           5.2          2.3
146           6.3          2.5           5.0          1.9
147           6.5          3.0           5.2          2.0
148           6.2          3.4           5.4          2.3
149           5.9          3.0           5.1          1.8

[100 rows x 4 columns]
```

[25]
```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=0.7,test_size = 0.3,random_state = 42)
```

[27]
```python
from sklearn.linear_model import LogisticRegression
classifier= LogisticRegression()
```

[34]
```python
from sklearn.model_selection import GridSearchCV
parameter={'penalty':['l1','l2','elasticnet'],'C':[1,2,3,4,5,6,10,20,30,40,50],'max_iter':[100,200,300,400]}
```

[41]
```python
classifier regressor=GridSearchCV(classifier,param grid=parameter,scoring='accuracy',cv=5)
```

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CONCLUSION:

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DISCUSSION AND VIVA VOCE:

1. When would you prefer Logistic Regression over complex classifiers like Random Forest or XGBoost in industry projects?
2. How do you interpret Logistic Regression coefficients in a business context?
3. How do you handle class imbalance when using Logistic Regression in real-world datasets?
4. Why is feature scaling important for Logistic Regression in production systems?
5. How do you choose the right threshold for classification in business applications?
6. How do evaluation metrics like precision, recall, ROC-AUC impact business decisions?
7. How do you prevent overfitting in Logistic Regression models used in industry?
8. What checks do you perform before deploying a Logistic Regression model to production?

REFERENCE:

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- https://medium.com/analytics-vidhya/20-interview-questions-on-linear-regression-and-logistic-regression-ef4d341d2805
- www.cs.sfu.ca/~han/MachineLearnig.html